5.2 BLE Proviosioning Procedure

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Terminologies:

EzloPi Device : EzloPi device is a controller device. It is developed based on ESP32 chipsets and has WiFi and BLE capabilities. It is also a controller when termed wrt Ezlo Cloud.

Onboardong : It is the procedure to share the WiFi credentials to EzloPi Device so that EzloPi device can further communicate to the internet.

Provisioning : It is the procedure to share all the info related to the Ezlo Cloud instance for an EzloPi Device. In other words it is the procedssure of sharing the info such as SSL Keys, CA certificates, UUIDs to the EzloPi device so that it can be further able to communite with Ezlo Cloud.

EzloPi firmware : It can be explained as the firmware that has only the logics to function but no any WiFi or provisioning info.

EzloPi UID : Each EzloPi device contains and UID, which is based on WiFi MAC and is being retained even if the flash is fully erased.

Central Device : Mobile device or Hub.

Description of the on-boarding process

EzloPi Power & Play

Classification of devices:

The application interacts with two types of devices: Controller and Hub/App.

1. Ezlopi Controller: is an Ezlopi device (i.e. esp32/esp32s3 based devices). The controller will be initialized as BLE-GATT-Server. It has four GATT-Services with different characteristics:
    1. 0x00E1 is used to configure and store wifi credentials
    2. 0x00E2 is used for factory reset
    3. 0x00E3 is used to store (cloud/manufacturer) provisioning data and, to fetch provisioning status
    4. 0x00E4 is used to get device information.
2. Hub/App:
   Hub is a device which can connect with the ezlopi-controllers. App is the android/ios/web/desktop application, which can connect with Ezlopi controllers. Hub/App will run as BLE-GATT-Client. It can connect with the Ezlopi controllers and can do read/write operations on it.

The process starts with the EzloPi device (also known as controller) sending advertising packets periodically at a predefined advertising interval. These packets contain the name of the device which depending if the device is provisioned or not (factory) will have a different format:

- Device provisioned is formatted as ezlopi_xxxx_yyyy (where xxxx is the ezlopi device type i.e. generic, ir_blaster, rf_blaster etc. and yyyy is the device serial.)
- Device not provisioned (factory/manufacturing scenario) is formatted as ezlopi_xxxxx (where xxxx is device type i.e. generic, ir_blaster, rf_blaster etc.).

The hub or the smartphone app (central role) can continuously scan for these advertising packets and when one is detected. Next the client (in our case app/hub) will send a connection request to the server (in our case Ezlopi).

Note: Ezlopi will stop advertising over BLE once a successful connection has been established which can be described as:

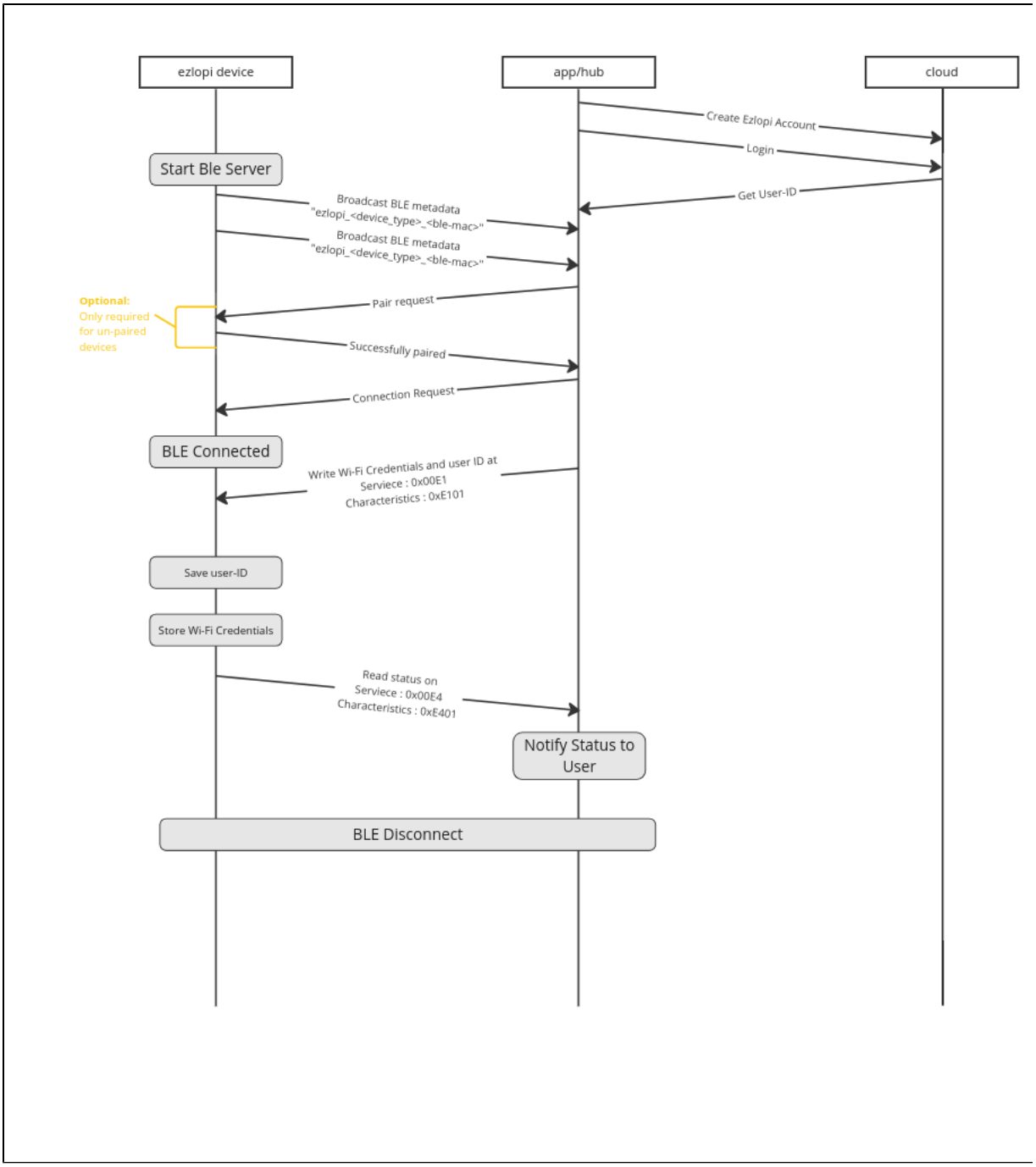
- Search the discovered device in paired list > connection request > connection success.

Next, the hub/app(client) will check the available services and characteristics on the newly connected Ezlopi device(Server).'

Use Cases

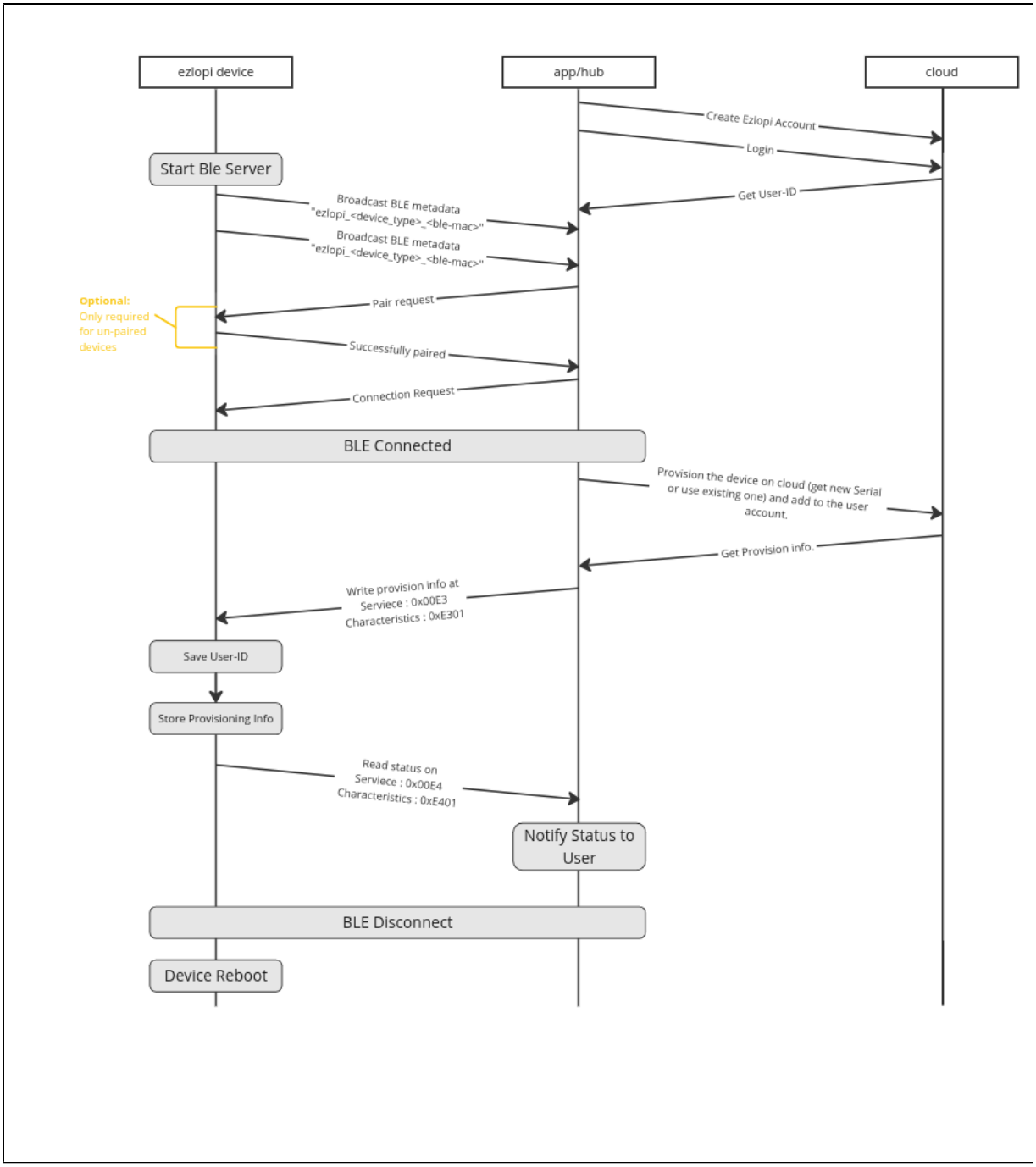
There are several use cases for both on-boarding and provisioning. It is to note that provisioning is only done after it has been on-boarded.

1. **New EzloPi Device (EzloPi device just containing EzloPi firmware)**
   This is the case when the EzloPi device contains just the EzloPi Firmware and no any WiFi and provisioning info. It can be termed as a fresh EzloPi device. This also indicates the EzloPi based sensor devices like presence sensor pro, IR/RF blaster etc. which has just been purchased by an user. It is necessary to onboard EzloPi device from this stage to make it functional. Below the on-boarding procedure is explained with a diagram.
   a. **On-boarding Procedure**



The figure above illustrates use cases when the device has never been on-boarded. EzloPi device as powers on it continuously broadcasts its packets over BLE. It can be discovered with name "ezlopi_<ble-mac>". First user using the central device (mobile device or hub) pairs to the ezlopi device and gets connected to the BLE server. Once connected to the BLE server, the central device can write WiFi credentials including User-ID. In this case EzloPi device does contains any User-ID and hence it stored the User-ID. After writing the credentials user can read the status with different characteristics but in the same GATT service. The GAAT service for writing and reading the status is explained below in detail.

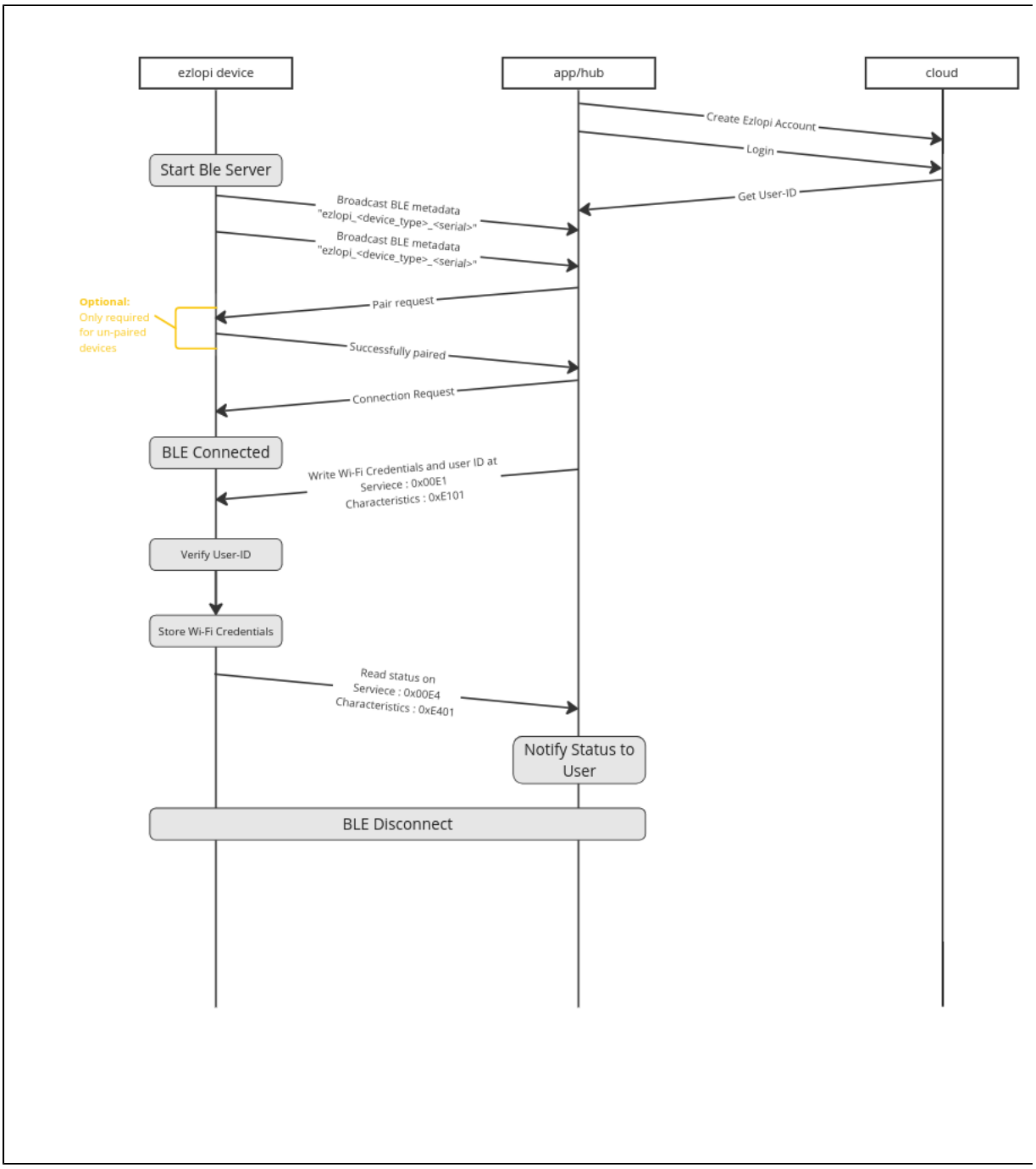
b. Provisioning Procedure



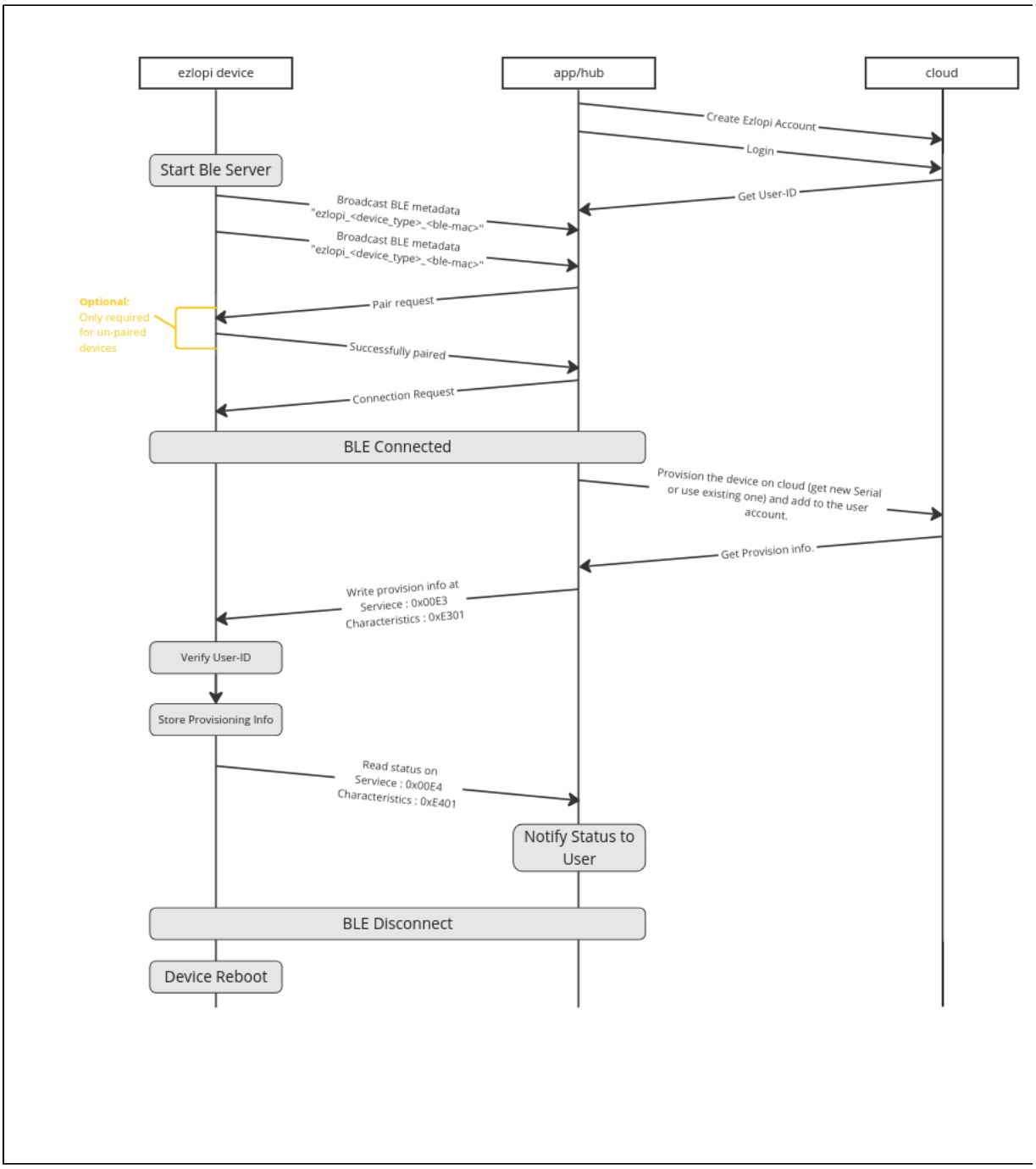
2. EzloPi Device already on-boarded and/or provisioned

EzloPi device has been on-boarded/provisioned or not can be read at anytime, the characteristics and service are explained at device info below. If the EzloPi device has been on-boarded or provisioned already we check the user info (user-id at the moment). This is to validate if the same user is changing the info or its an intruder. The on-boarded and provisioned device in the BLE broadcasts its name as **ezlopi_<device_type>_<device_serial>**. The flow for it is presented with the diagram below :

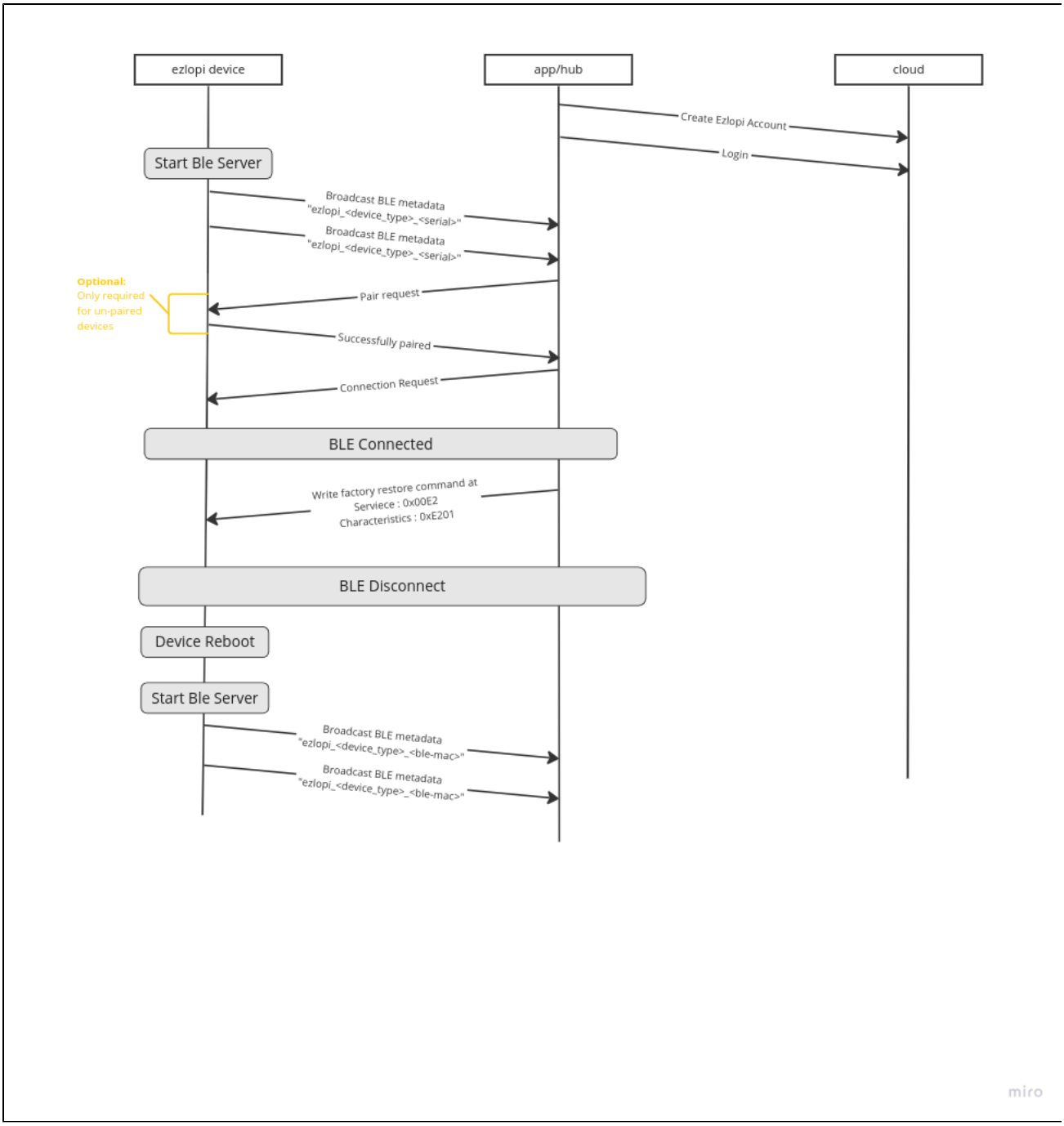
a. On-boarding Procedure



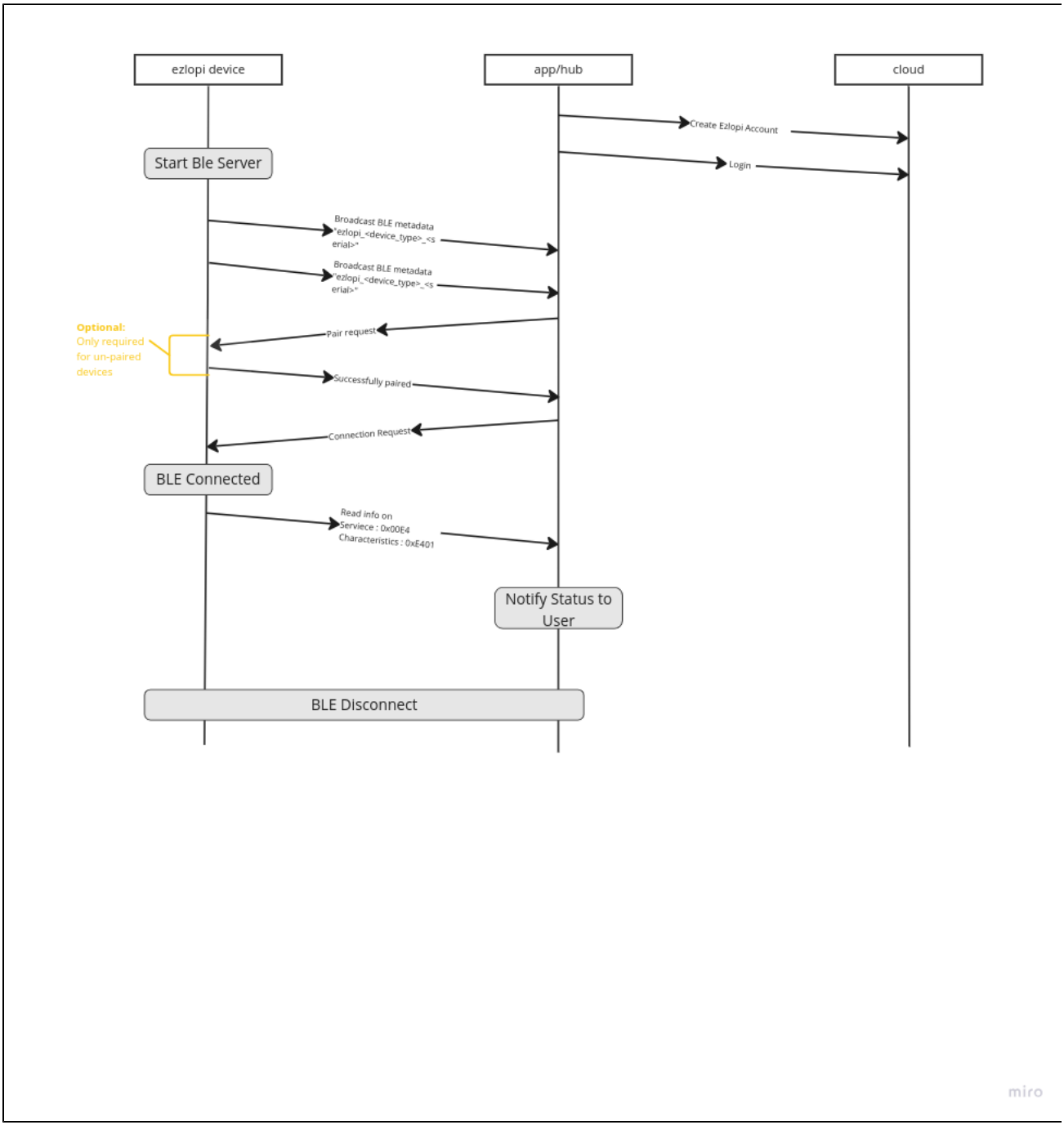
b. Provisioning Procedure



3. **EzloPi Device factory restore**


ezlopi device
app/hub
cloud
Create Ezlopi Account
Login
Start Ble Server
Broadcast BLE metadata
"ezlopi_<device_type>_<serial>"
Broadcast BLE metadata
"ezlopi_<device_type>_<serial>"
Optional:
Only required
for un-paired
devices
Pair request
Successfully paired
Connection Request
BLE Connected
Write factory restore command at
Serviece : 0x00E2
Characteristics : 0xE201
BLE Disconnect
Device Reboot
Start Ble Server
Broadcast BLE metadata
"ezlopi_<device_type>_<ble-mac>"
Broadcast BLE metadata
"ezlopi_<device_type>_<ble-mac>"
miro

4. **EzloPi Device info read**



Services and Characteristics Descriptions:

1. **WiFi Service *(Service UUID: 0x00E1)*:**

   Wi-Fi Service is only for writing Wi-Fi configuration and reading its status.

   a. **WiFi Credentials *(Characteristics UUID: 0xE101)*:**

      This characteristics is used to configure the WiFi to the controller. BLE hub/app can both read and write the value in json format.
      Sending write command to the controller:

WiFi Write [S: 0x00E1, C: 0xE101]

```
{
        "wifi_ssid": "WiFi SSID",
        "wifi_password": "password",
        "user_id": "Jon"
}
```

2. ## Factory Reset Service *(UUID: 0x00E2)*:

Factory reset service is currently being used only for sending factory reset command. The controller will erase all the written data from flash as well as nvs-flash.

a. Factory Reset *(Characteristics UUID: 0xE202)*:

Deprecated since 3.4.7

This characteristics deprecated since version 3.4.7 and will be removed in future versions.

This characteristics is used to issue factory reset command to the controller. It is write only characteristics. It means hub/app can only write through this characteristics but can not read. The device will restart after issuing this command. The command is in json format.
Write command:

Factory restore command

```
{
        "cmd": 2
}
```

b. Reset *(Characteristics UUID: 0xE203)*:

Reset Command added on V3.4.7

This command has been added since the version V3.4.7

This characteristics is used to issue reboot device or factory restore

EzloPi Command 0, Reset

```
{
    "cmd": 0,
    "sub_cmd": 0
}
```

Key – Value description

S.N.	Key	Details	Type	Data
Request From UI				
1	cmd	Command Code/ID	Number	0: Reset
2	sub_cmd	Sub command to main command	Number	0: Factory Restore 1: Reboot Device

3. ## Provisioning Service *(Service UUID: 0x00E3)*:

a. Provisioning Info *(Characteristics UUID: 0xE301)*:

This characteristics is for writing provisioning data to the controller. Provisioning data mainly consists of cloud information such as certificates, endpoint, uuid, serial-number, etc., and manufacturer information such as model, type, manufacturer-name, etc. Since it contains critical information such like certificates, the characteristics is kept as write only. The provisioning information needs to be in json format.
Write example:

Provisioning info

```
{
  "serial": 100004961,
  "provision_server": "https://req-disp-at0m.mios.com/",
  "cloud_server": "https://cloud.ezlo.com:7000",
```

```
  "provision_token":
"c25415a5f1de9e5084dab5b0c92a28d2f5f64bb0079a9ffcc312eb005ff1d2b9bb896a097a565eb79fb15334f7c296e8ca
67596cde8a87255d44e2507face3c288aed92624e355207700b07071365353acca8a1c3e6ac2e5dcce519021d2bc72cc7f5
c4e85ebd0e75cc84c8b56a8a378672bcd9ac6975b511b77482001ab9a35",
  "ssl_private_key": "-----BEGIN PRIVATE KEY-----\nMIGEAgEAMBAGByqGSM49AgEGBSuBBAAKBG0wawIBAQQgpADG
/nseS5R6XDv15FcM\ntU9zNVVal8X9qzESKrLzWDehRANCAATPmLXLTDUTxBvNIm6BFhzmGirTmW+wBE8B\nAEJJb1c9zuSdSMd
MMUeb89KGb3scQUZotkiH/ki/VR0+3Lvorg8v\n-----END PRIVATE KEY-----\n",
  "ssl_public_key": "-----BEGIN PUBLIC KEY-----
\nMFYwEAYHKoZIzj0CAQYFK4EEAAoDQgAEz5i1y0w1E8QbzSJugRYc5hoq05lvsARP\nAQBCSW9XPc7knUjHTDFHm
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  "ssl_shared_key": "\n-----BEGIN CERTIFICATE-----
\nMIICDTCCAbKgAwIBAgIDAzZDMAoGCCqGSM49BAMCMIGQMQswCQYDVQQGEwJVUzEU\nMBIGA1UECAwLIE5ldyBKZXJzZXkxEDA
OBgNVBAcMB0NsaWZ0b24xDzANBgNVBAoM\nBklUIE9wczEPMA0GA1UECwwGSVQgT3BzMRQwEgYDVQQDDAtlWkxPIExURCBDQTEh
\nMB8GCSqGSIb3DQEJARYSc3lzYWRtaW5zQGV6bG8uY29tMCAXDTIxMTIwMjA4NDYx\nMFoYDzIyOTUwOTE2MDg0NjEwWjCBjDE
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\nLTArBgNVBAsMJDRjMGQ5MDYwLTUzNGMtMTFlYy1iMmQ2LThmMjYwZjUyODdmYTES\nMBAGA1UEAwwJMTAwMDA0OTYxMFYwEAY
HKoZIzj0CAQYFK4EEAAoDQgAEz5i1y0w1\nE8QbzSJugRYc5hoq05lvsARPAQBCSW9XPc7knUjHTDFHm/PShm97HEFGaLZIh
/5I\nv1UdPty76K4PLzAKBggqhkjOPQQDAgNJADBGAiEAseX9mwtQ2DN60jCplOIIPd2S\n+6bdgpQGiCeYNpY
/sBcCIQDwduNJYOIlBGmQgH1KAk9nW5JxAzA2MsPKovjTUSFB\nyg==\n-----END CERTIFICATE-----\n",
  "signing_ca_certificate": "-----BEGIN CERTIFICATE-----
\r\nMIICbDCCAhGgAwIBAgIJAOByzaI7aHY9MAoGCCqGSM49BAMDMIGQMQswCQYDVQQG\r\nEwJVUzEUMBIGA1UECAwLIE5ldyB
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xPIExU\r\nRCBDQTEhMB8GCSqGSIb3DQEJARYSc3lzYWRtaW5zQGV6bG8uY29tMCAXDTE5MDUz\r\nMTE3MDE0N1oYDzIxMTkwN
TA3MTcwMTQ3WjCBkDELMAkGA1UEBhMCVVMxFDASBgNV\r\nBAgMCyBOZXcgSmVyc2V5MRAwDgYDVQQHDAdDbGlmdG9uMQ8wDQYD
VQQKDAZJVCBP\r\ncHMxDzANBgNVBAsMBklUIE9wczEUMBIGA1UEAwwLZVpMTyBMVEQgQ0ExITAfBgkq\r\nhkiG9w0BCQEWEnN
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fBgNVHSMEGDAWgBRiTl8Ez1l94jaqcxbiyxkVC0FkBTAPBgNVHRMB\r\nAf8EBTADAQH
/MAoGCCqGSM49BAMDA0kAMEYCIQD7EUs8j50jKFd
/46Zo95NbrPYQ\r\nPtLTHH9YjUkMEkYD5gIhAMP4y7E1aB78nQrmd3IX8MM32k9dM8xT0MztR16OtsuV\r\n-----END
CERTIFICATE-----",
  "uuid": "4c0d9060-534c-11ec-b2d6-8f260f5287fa",
  "user_id": "Jon",
  "device_name": "My Device",
  "brand": "doe",
  "manufacturer_name": "foo",
  "model_number": "063DEX524",
  "mac": "11:22:33:44:55:66",
  "uuid_provisioning": "b558c598-a35a-4fdc-a2cd-5462ff11455a",
  "device_type_ezlopi": "ezlopi_generic"
}
```


Before sharing the provisioning info via BLE we minify and encode it into bsae64 characters.
The minified JSON can be as :

Provisioning info : Minified JSON


```
{"serial":100004961,"provision_server":"https://req-disp-at0m.mios.com/","cloud_server":"
https://cloud.ezlo.com:7000","provision_token":"
c25415a5f1de9e5084dab5b0c92a28d2f5f64bb0079a9ffcc312eb005ff1d2b9bb896a097a565eb79fb15334f7c296e8ca6
7596cde8a87255d44e2507face3c288aed92624e355207700b07071365353acca8a1c3e6ac2e5dcce519021d2bc72cc7f5c
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PUBLIC KEY-----
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/PShm97HEFGaLZIh/5Iv1UdPty76K4PLw==\n-----END PUBLIC KEY-----","ssl_shared_key":"\n-----BEGIN
CERTIFICATE-----
\nMIICDTCCAbKgAwIBAgIDAzZDMAoGCCqGSM49BAMCMIGQMQswCQYDVQQGEwJVUzEU\nMBIGA1UECAwLIE5ldyBKZXJzZXkxEDA
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```

```
signing_ca_certificate":"-----BEGIN CERTIFICATE-----
\r\nMIICbDCCAhGgAwIBAgIJAOByzaI7aHY9MAoGCCqGSM49BAMDMIGQMQswCQYDVQQG\r\nEwJVUzEUMBIGA1UECAwLIE5ldyB
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/46Zo95NbrPYQ\r\nPtLTHH9YjUkMEkYD5gIhAMP4y7E1aB78nQrmd3IX8MM32k9dM8xT0MztR16OtsuV\r\n-----END
CERTIFICATE-----","uuid":"4c0d9060-534c-11ec-b2d6-8f260f5287fa","user_id":"Jon","device_name":"My
Device","brand":"doe","manufacturer_name":"foo","model_number":"063DEX524","mac":"11:22:33:44:55:
66","uuid_provisioning":"b558c598-a35a-4fdc-a2cd-5462ff11455a","device_type_ezlopi":"generic"}
```


The minified JSON is then converted to Base64 encoded string.

Provisioning info : base64 encoded


```
eyJzZXJpYWwiOjEwMDAwNDk2MSwicHJvdmlzaW9uX3NlcnZlciI6Imh0dHBzOi8vcmVxLWRpc3AtYXQwbS5taW9zLmNvbS8iLCJ
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zM0ZjdjMjk2ZThjYTY3NTk2Y2RlOGE4NzI1NWQ0NGUyNTA3ZmFjZTNjMjg4YWVkOTI2MjRlMzU1MjA3NzAwYjA3MDcxMzY1MzUz
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DlxekVTS3JMeldEZWhSQU5DQUFUUG1MWExURFVUeEJ2TkltNkJGaHptR2lyVG1XK3dCRThCXG5BRUpKYjFjOXp1U2RTTWRNTVVl
Yjg5S0diM3NjUVVab3RraUgva2kvVlIwKzNMdm9yZzh2XG4tLS0tLUVORCBQUklWQVRFIEtFWS0tLS0tXG4iLCJzc2xfcHVibGl
jX2tleSI6Ii0tLS0tQkVHSU4gUFVCTElDIEtFWS0tLS0tXG5NRll3RUFZSEtvWkl6ajBDQVFZRks0RUVBQW9EUWdBRXo1aTF5MH
cxRThRYnpTSnVnUlljNWhvcTA1bHZzQVJQXG5BUUJDU1c5WFBjN2tuVWpIVERGSG0vUFNobTk3SEVGR2FMWkloLzVJdjFVZFB0e
Tc2SzRQTHc9PVxuLS0tLS1FTkQgUFVCTElDIEtFWS0tLS0tIiwic3NsX3NoYXJlZF9rZXkiOiJcbi0tLS0tQkVHSU4gQ0VSVElG
SUNBVEUtLS0tLVxuTUlJQ0RUQ0NBYktnQXdJQkFnSURBelpETUFvR0NDcUdTTTQ5QkFNQ01JR1FNUXN3Q1FZRFZRUUdFd0pWVXp
FVVxuTUJJR0ExVUVDQXdMSUU1bGR5QktaWEp6WlhreEVEQU9CZ05WQkFjTUIwTnNhV1owYjI0eER6QU5CZ05WQkFvTVxuQmtsVU
lFOXdjekVQTUEwR0ExVUVDd3dHU1ZRZ1QzQnpNUlF3RWdZRFZRUUREQXRsV2t4UElFeFVSQ0JEUVRFaFxuTUI4R0NTcUdTSWIzR
FFFSkFSWVNjM2x6WVdSdGFXNXpRR1Y2Ykc4dVkyOXRNQ0FYRFRJeE1USXdNakE0TkRZeFxuTUZvWUR6SXlPVFV3T1RFMk1EZzBO
akV3V2pDQmpERUxNQWtHQTFVRUJoTUNWVk14RXpBUkJnTlZCQWdNQ2s1bFxuZHlCS1pYSnpaWGt4RURBT0JnTlZCQWNNQjBOc2F
XWjBiMjR4RXpBUkJnTlZCQW9NQ21OdmJuUnliMnhzWlhJeFxuTFRBckJnTlZCQXNNSkRSak1HUTVNRFl3TFRVek5HTXRNVEZsWX
kxaU1tUTJMVGhtTWpZd1pqVXlPRGRtWVRFU1xuTUJBR0ExVUVBd3dKTVRBd01EQTBPVFl4TUZZd0VBWUhLb1pJemowQ0FRWUZLN
EVFQUFvRFFnQUV6NWkxeTB3MVxuRThRYnpTSnVnUlljNWhvcTA1bHZzQVJQQVFCQ1NXOVhQYzdrblVqSFRERkhtL1BTaG05N0hF
RkdhTFpJaC81SVxudjFVZFB0eTc2SzRQTHpBS0JnZ3Foa2pPUFFRREFnTkpBREJHQWlFQXNlWDltd3RRMkRONjBqQ3BsT0lJUGQ
yU1xuKzZiZGdwUUdpQ2VZTnBZL3NCY0NJUUR3ZHVOSllPSWxCR21RZ0gxS0FrOW5XNUp4QXpBMk1zUEtvdmpUVVNGQlxueWc9PV
xuLS0tLS1FTkQgQ0VSVElGSUNBVEUtLS0tLVxuIiwic2lnbmluZ19jYV9jZXJ0aWZpY2F0ZSI6Ii0tLS0tQkVHSU4gQ0VSVElGS
UNBVEUtLS0tLVxyXG5NSUlDYkRDQ0FoR2dBd0lCQWdJSkFPQnl6YUk3YUhZOU1Bb0dDQ3FHU000OUJBTURNSUdRTVFzd0NRWURW
UVFHXHJcbkV3SlZVekVVTUJJR0ExVUVDQXdMSUU1bGR5QktaWEp6WlhreEVEQU9CZ05WQkFjTUIwTnNhV1owYjI0eER6QU5cclx
uQmdOVkJBb01Ca2xVSUU5d2N6RVBNQTBHQTFVRUN3d0dTVlFnVDNCek1SUXdFZ1lEVlFRRERBdGxXa3hQSUV4VVxyXG5SQ0JEUV
RFaE1COEdDU3FHU0liM0RRRUpBUllTYzNsellXUnRhVzV6UUdWNmJHOHVZMjl0TUNBWERURTVNRFV6XHJcbk1URTNNREUwTjFvW
UR6SXhNVGt3TlRBM01UY3dNVFEzV2pDQmtERUxNQWtHQTFVRUJoTUNWVk14RkRBU0JnTlZcclxuQkFnTUN5Qk9aWGNnU21WeWMy
VjVNUkF3RGdZRFZRUUhEQWREYkdsbWRHOXVNUTh3RFFZRFZRUUtEQVpKVkNCUFxyXG5jSE14RHpBTkJnTlZCQXNNQmtsVUlFOXd
jekVVTUJJR0ExVUVBd3dMWlZwTVR5Qk1WRVFnUTBFeElUQWZCZ2txXHJcbmhraUc5dzBCQ1FFV0VuTjVjMkZrYldsdWMwQmxlbX
h2TG1OdmJUQldNQkFHQnlxR1NNNDlBZ0VHQlN1QkJBQUtcclxuQTBJQUJITTFFkaExEWXNhZklGWThwWmg5NmFER3FWbTZFNHI4b
lc5czRDZmRwWGFhL1I0Q25qYVZwRFFJN1VtUVxyXG45dlZER1puOG1jbW03VmpLeCtUU0NTME1JS09qVXpCUk1CMEdBMVVkRGdR
V0JCUmlUbDhFejFsOTRqYXFjeGJpXHJcbnl4a1ZDMEZrQlRBZkJnTlZIU01FR0RBV2dCUmlUbDhFejFsOTRqYXFjeGJpeXhrVkM
wRmtCVEFQQmdOVkhSTUJcclxuQWY4RUJUQURBUUgvTUFvR0NDcUdTTTQ5QkFNREEwa0FNRVlDSVFEN0VVczhqNTBqS0ZkLzQ2Wm
85NU5iclBZUVxyXG5QdExUSEg5WWpVa01Fa1lENWdJaEFNUDR5N0UxYUI3OG5Rcm1kM0lYOE1NMzJrOWRNOHhUME16dFIxNk90c
3VWXHJcbi0tLS0tRU5EIENFUlRJRklDQVRFLS0tLS0iLCJ1dWlkIjoiNGMwZDkwNjAtNTM0Yy0xMWVjLWIyZDYtOGYyNjBmNTI4
N2ZhIiwidXNlcl9pZCI6IkpvbiIsImRldmljZV9uYW1lIjoiTXkgRGV2aWNlIiwiYnJhbmQiOiJkb2UiLCJtYW51ZmFjdHVyZXJ
fbmFtZSI6ImZvbyIsIm1vZGVsX251bWJlciI6IjA2M0RFWDUyNCIsIm1hYyI6IjExOjIyOjMzOjQ0OjU1OjY2IiwidXVpZF9wcm
92aXNpb25pbmciOiJiNTU4YzU5OC1hMzVhLTRmZGMtYTJjZC01NDYyZmYxMTQ1NWEiLCJkZXZpY2VfdHlwZV9lemxvcGkiOiJnZ
W5lcmljIn0=
```


This base64 characters then sent with 400 characters each time as :

Single chunk of data to be sent.

```
{
  "total_len": 3872, // Total length of base64 string
  "len": 400,                // Length of the chunk
  "sequence": 1,         // data sequence number
  "data":
"eyJzZXJpYWwiOjEwMDAwNDk2MSwicHJvdmlzaW9uX3NlcnZlciI6Imh0dHBzOi8vcmVxLWRpc3AtYXQwbS5taW9zLmNvbS8iLC
JjbG91ZF9zZXJ2ZXIiOiJodHRwczovL2Nsb3VkLmV6bG8uY29tOjcwMDAiLCJwcm92aXNpb25fdG9rZW4iOiJjMjU0MTVhNWYxZ
GU5ZTUwODRkYWI1YjBjOTJhMjhkMmY1ZjY0YmIwMDc5YTlmZmNjMzEyZWIwMDVmZjFkMmI5YmI4OTZhMDk3YTU2NWViNzlmYjE1
MzM0ZjdjMjk2ZThjYTY3NTk2Y2RlOGE4NzI1NWQ0NGUyNTA3ZmFjZTNjMjg4YWVkOTI2MjRlMzU1MjA3NzAwYjA3MDcxMzY1MzU
zYWNj"
}
```

The whole data can be represented as :

Multiple sequence of base64 message.

```
{"total_len":3872,"len":400,"sequence":1,"data":"
eyJzZXJpYWwiOjEwMDAwNDk2MSwicHJvdmlzaW9uX3NlcnZlciI6Imh0dHBzOi8vcmVxLWRpc3AtYXQwbS5taW9zLmNvbS8iLCJ
jbG91ZF9zZXJ2ZXIiOiJodHRwczovL2Nsb3VkLmV6bG8uY29tOjcwMDAiLCJwcm92aXNpb25fdG9rZW4iOiJjMjU0MTVhNWYxZG
U5ZTUwODRkYWI1YjBjOTJhMjhkMmY1ZjY0YmIwMDc5YTlmZmNjMzEyZWIwMDVmZjFkMmI5YmI4OTZhMDk3YTU2NWViNzlmYjE1M
zM0ZjdjMjk2ZThjYTY3NTk2Y2RlOGE4NzI1NWQ0NGUyNTA3ZmFjZTNjMjg4YWVkOTI2MjRlMzU1MjA3NzAwYjA3MDcxMzY1MzUz
YWNj"}
{"total_len":3872,"len":400,"sequence":2,"data":"
YThhMWMzZTZhYzJlNWRjY2U1MTkwMjFkMmJjNzJjYzdmNWM0ZTg1ZWJkMGU3NWNjODRjOGI1NmE4YTM3ODY3MmJjZDlhYzY5NzV
iNTExYjc3NDgyMDAxYWI5YTM1Iiwic3NsX3ByaXZhdGVfa2V5IjoiLS0tLS1CRUdJTiBQUklWQVRFIEtFWS0tLS0tXG5NSUdFQW
dFQU1CQUdCeXFHU000OUFnRUdCU3VCQkFBS0JHMHdhd0lCQVFRZ3BBREcvbnNlUzVSNlhEdjE1RmNNXG50VTl6TlZWYWw4WDlxe
kVTS3JMeldEZWhSQU5DQUFUUG1MWExURFVUeEJ2TkltNkJGaHptR2lyVG1XK3dCRThCXG5BRUpKYjFjOXp1U2RTTWRNTVVlYjg5
S0di"}
{"total_len":3872,"len":400,"sequence":3,"data":"
M3NjUVVab3RraUgva2kvVlIwKzNMdm9yZzh2XG4tLS0tLUVORCBQUklWQVRFIEtFWS0tLS0tXG4iLCJzc2xfcHVibGljX2tleSI
6Ii0tLS0tQkVHSU4gUFVCTElDIEtFWS0tLS0tXG5NRll3RUFZSEtvWkl6ajBDQVFZRks0RUVBQW9EUWdBRXo1aTF5MHcxRThRYn
pTSnVnUlljNWhvcTA1bHZzQVJQXG5BUUJDU1c5WFBjN2tuVWpIVERGSG0vUFNobTk3SEVGR2FMWkloLzVJdjFVZFB0eTc2SzRQT
Hc9PVxuLS0tLS1FTkQgUFVCTElDIEtFWS0tLS0tIiwic3NsX3NoYXJlZF9rZXkiOiJcbi0tLS0tQkVHSU4gQ0VSVElGSUNBVEUt
LS0t"}
{"total_len":3872,"len":400,"sequence":4,"data":"
LVxuTUlJQ0RUQ0NBYktnQXdJQkFnSURBelpETUFvR0NDcUdTTTQ5QkFNQ01JR1FNUXN3Q1FZRFZRUUdFd0pWVXpFVVxuTUJJR0E
xVUVDQXdMSUU1bGR5QktaWEp6WlhreEVEQU9CZ05WQkFjTUIwTnNhV1owYjI0eER6QU5CZ05WQkFvTVxuQmtsVUlFOXdjekVQTU
EwR0ExVUVDd3dHU1ZRZ1QzQnpNUlF3RWdZRFZRUUREQXRsV2t4UElFeFVSQ0JEUVRFaFxuTUI4R0NTcUdTSWIzRFFFSkFSWVNjM
2x6WVdSdGFXNXpRR1Y2Ykc4dVkyOXRNQ0FYRFRJeE1USXdNakE0TkRZeFxuTUZvWUR6SXlPVFV3T1RFMk1EZzBOakV3V2pDQmpE
RUxN"}
{"total_len":3872,"len":400,"sequence":5,"data":"
QWtHQTFVRUJoTUNWVk14RXpBUkJnTlZCQWdNQ2s1bFxuZHlCS1pYSnpaWGt4RURBT0JnTlZCQWNNQjBOc2FXWjBiMjR4RXpBUkJ
nTlZCQW9NQ21OdmJuUnliMnhzWlhJeFxuTFRBckJnTlZCQXNNSkRSak1HUTVNRFl3TFRVek5HTXRNVEZsWXkxaU1tUTJMVGhtTW
pZd1pqVXlPRGRtWVRFU1xuTUJBR0ExVUVBd3dKTVRBd01EQTBPVFl4TUZZd0VBWUhLb1pJemowQ0FRWUZLNEVFQUFvRFFnQUV6N
WkxeTB3MVxuRThRYnpTSnVnUlljNWhvcTA1bHZzQVJQQVFCQ1NXOVhQYzdrblVqSFRERkhtL1BTaG05N0hFRkdhTFpJaC81SVxu
djFV"}
{"total_len":3872,"len":400,"sequence":6,"data":"
ZFB0eTc2SzRQTHpBS0JnZ3Foa2pPUFFRREFnTkpBREJHQWlFQXNlWDltd3RRMkRONjBqQ3BsT0lJUGQyU1xuKzZiZGdwUUdpQ2V
ZTnBZL3NCY0NJUUR3ZHVOSllPSWxCR21RZ0gxS0FrOW5XNUp4QXpBMk1zUEtvdmpUVVNGQlxueWc9PVxuLS0tLS1FTkQgQ0VSVE
lGSUNBVEUtLS0tLVxuIiwic2lnbmluZ19jYV9jZXJ0aWZpY2F0ZSI6Ii0tLS0tQkVHSU4gQ0VSVElGSUNBVEUtLS0tLVxyXG5NS
UlDYkRDQ0FoR2dBd0lCQWdJSkFPQnl6YUk3YUhZOU1Bb0dDQ3FHU000OUJBTURNSUdRTVFzd0NRWURWUVFHXHJcbkV3SlZVekVV
TUJJ"}
{"total_len":3872,"len":400,"sequence":7,"data":"
R0ExVUVDQXdMSUU1bGR5QktaWEp6WlhreEVEQU9CZ05WQkFjTUIwTnNhV1owYjI0eER6QU5cclxuQmdOVkJBb01Ca2xVSUU5d2N
6RVBNQTBHQTFVRUN3d0dTVlFnVDNCek1SUXdFZ1lEVlFRRERBdGxXa3hQSUV4VVxyXG5SQ0JEUVRFaE1COEdDU3FHU0liM0RRRU
pBUllTYzNsellXUnRhVzV6UUdWNmJHOHVZMjl0TUNBWERURTVNRFV6XHJcbk1URTNNREUwTjFvWUR6SXhNVGt3TlRBM01UY3dNV
FEzV2pDQmtERUxNQWtHQTFVRUJoTUNWVk14RkRBU0JnTlZcclxuQkFnTUN5Qk9aWGNnU21WeWMyVjVNUkF3RGdZRFZRUUhEQWRE
Ykds"}
{"total_len":3872,"len":400,"sequence":8,"data":"
bWRHOXVNUTh3RFFZRFZRUUtEQVpKVkNCUFxyXG5jSE14RHpBTkJnTlZCQXNNQmtsVUlFOXdjekVVTUJJR0ExVUVBd3dMWlZwTVR
5Qk1WRVFnUTBFeElUQWZCZ2txXHJcbmhraUc5dzBCQ1FFV0VuTjVjMkZrYldsdWMwQmxlbXh2TG1OdmJUQldNQkFHQnlxR1NNND
lBZ0VHQlN1QkJBQUtcclxuQTBJQUJITTFFkaExEWXNhZklGWThwWmg5NmFER3FWbTZFNHI4blc5czRDZmRwWGFhL1I0Q25qYVZwR
FFJN1VtUVxyXG45dlZER1puOG1jbW03VmpLeCtUU0NTME1JS09qVXpCUk1CMEdBMVVkRGdRV0JCUmlUbDhFejFsOTRqYXFjeGJp
XHJc"}
{"total_len":3872,"len":400,"sequence":9,"data":"
bnl4a1ZDMEZrQlRBZkJnTlZIU01FR0RBV2dCUmlUbDhFejFsOTRqYXFjeGJpeXhrVkMwRmtCVEFQQmdOVkhSTUJcclxuQWY4RUJ
UQURBUUgvTUFvR0NDcUdTTTQ5QkFNREEwa0FNRVlDSVFEN0VVczhqNTBqS0ZkLzQ2Wm85NU5iclBZUVxyXG5QdExUSEg5WWpVa0
1Fa1lENWdJaEFNUDR5N0UxYUI3OG5Rcm1kM0lYOE1NMzJrOWRNOHhUME16dFIxNk90c3VWXHJcbi0tLS0tRU5EIENFUlRJRklDQ
```

```
VRFLS0tLS0iLCJ1dWlkIjoiNGMwZDkwNjAtNTM0Yy0xMWVjLWIyZDYtOGYyNjBmNTI4N2ZhIiwidXNlcl9pZCI6IkpvbiIsImRl
dmlj"}
{"total_len":3872,"len":272,"sequence":10,"data":"
ZV9uYW1lIjoiTXkgRGV2aWNlIiwiYnJhbmQiOiJkb2UiLCJtYW51ZmFjdHVyZXJfbmFtZSI6ImZvbyIsIm1vZGVsX251bWJlciI
6IjA2M0RFWDUyNCIsIm1hYyI6IjExOjIyOjMzOjQ0OjU1OjY2IiwidXVpZF9wcm92aXNpb25pbmciOiJiNTU4YzU5OC1hMzVhLT
RmZGMtYTJjZC01NDYyZmYxMTQ1NWEiLCJkZXZpY2VfdHlwZV9lemxvcGkiOiJnZW5lcmljIn0="}
```


4. **Device Info *(Service UUID: 0x00E4)*:**
   a. **Device Info *(Characteristics UUID: 0xE401)*:**

Deprecated since 3.4.7

This characteristics deprecated since version 3.4.7 and will be removed in future versions.

The EzloPi device (BLE GATT Server) broadcasts the info JSON at the service 0x00E4 and characteristics 0xE401. This is the read only characteristics. With the info in the characteristics, app can decide whether to provision the device or not.

EzloPi Device Info

```
{
  "firmware_version": "2.0.9",
  "firmware_build": 169,
  "chip": "esp32s3",
  "uptime": 113,                                          // seconds
  "build_date": 1690548178,                             // epoch
  "provisioned_status": true,
  "mac": "11:22:33:44:55:66",
  "serial": 100004961,
  "ezlopi_device_type": "ezlopi_generic",
  "wifi_ssid": "nepaldigisys",
  "wifi-connection_status": 1,                        // 0 = Not connected to WiFi, 1 = Connected
to WiFi
  "internet_status": 1                                    // 0 = Internet not Connected, 1 =
Internet not Connected
}
```

Newly added since 3.4.7

The following characteristics from 0xE402 - 0xE40A are newly added since version 3.4.7

   b. **API Version Info [Characteristics UUID: 0xE402] :**

E402 : APi Info

```
{
    "api": "1.0.0"
}
```

   c. **EzloPi Firmware Info [Characteristics UUID: 0xE403] :**

E402 : APi Info

```
{
    "version": "3.4.7",
    "build": 185,
    "build_date": "2024-02-29T18:46:33+545"
}
```

   d. **Microcontroller (Chip) Info [Characteristics UUID: 0xE404] :**

E402 : APi Info

```
{
    "type": "ESP32",
    "version": "1.00"
}
```

e. **Microcontroller HAL SDK/API info provided by the manufacturer [Characteristics UUID: 0xE405] :**

E402 : APi Info


```
{
    "name": "ESP-IDF",
    "version": "v4.4.6-dirty"
}
```


f. **EzloPi Device Current Status [Characteristics UUID: 0xE406] :**

E402 : APi Info


```
{
    "uptime": "0d 0h 1m 51s",
    "boot_count": 13,
    "boot_reason": "Power ON",
    "flash_size": "4MB",
    "wifi_mac": "EC:62:60:A7:1D:98",
    "ble_mac": "EC:62:60:A7:1D:9A"
}
```


g. **Serial Port Configuration [Characteristics UUID: 0xE407] :**

E402 : APi Info


```
{
    "baud": 115200,
    "parity": "N",
    "start_bits": 0,
    "stop_bits": 1,
    "frame_size": 8,
    "flow_control": "DISABLED"
}
```


h. **Ezlo Cloud related info for EzloPi [Characteristics UUID: 0xE408] :**

E402 : APi Info


```
{
    "provisioned": true,
    "uuid": "1b576400-8392-11ee-8889-0f56091e7639",
    "uuid_prov": "c2ea484b-6b31-446e-b42e-8ab7b5c36ce4",
    "type": "ezlopi_generic",
    "serial": 102002167,
    "mac": "a2:f8:75:1c:3a:4c"
}
```


i. **OEM related info [Characteristics UUID: 0xE409] :**

E402 : APi Info


```
{
    "brand": "",
    "manufacturer": "",
    "model": ""
}
```


j. **Current network related info on EzloPi [Characteristics UUID: 0xE40A] :**

E402 : APi Info


```
{
    "ssid": "nepaldigisys",
    "wifi_mode": "STA",
    "ip_sta": "192.168.1.96",
    "ip_nmask": "255.255.255.0",
    "ip_gw": "192.168.1.254",
    "wifi": true,
    "internet": true,
    "cloud": true
}
```